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North Karanpura Mine Performance Report

Opencast Expansion, Bench Stability Radar & Fleet Telematics

CATEGORY	MINE / BLOCK	YEAR	CLASSIFICATION
Mining	North Karanpura	2025	VERIFIED

DOCUMENT VERIFICATION & ACCESS REGISTER

Lead Author / Division: Er. Arunav Sen (Mining Engineer)

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Standard Operating Procedure: SOP-CI-MIN-2025-A

System Node: CLOUD-KNOWLEDGE-NODE-EAST-01

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SECTION 1.0 â€” DOCUMENT CONTROL & MINE LICENSE REGISTER

Official Technical Record & Geological Data Logging

Mining Operations Report covering quarterly extraction, bench development, heavy equipment logistics, and slope stability at Central West Mine.

Operations conducted under Mine Lease No. ML-CWS-098 and Environmental Clearance No. J-11015/284/2018-IA.II(M).

Statutory Mining Approvals Register

Approval / License	Authority / Directorate	Validity Period	Current Status
Mine Lease ML-098	Ministry of Coal	Valid to 2045	Active & Good Standing
DGMS Safety Clearance	Directorate Gen. Mines Safety	Annual Inspection	Compliant
Consolidated Consent (CTO)	State Pollution Control Board	Valid to 2028	Compliant
Explosive Storage License	PESO Nagpur	Valid to 2027	Approved Mag. #3

SECTION 2.0 â€” EXECUTIVE OPERATIONAL SUMMARY

Official Technical Record & Geological Data Logging

Central West Mine sustained an average quarterly production of 1.48 MT of Grade G-4 non-coking coal with an overburden handling volume of 4.22 M.Cu.m.

The box cut advance on Bench #3 proceeded strictly along the geological strike line, mitigating floor heaving hazards.

CRITICAL TECHNICAL OBSERVATION & DIRECTIVE:

Zero lost-time injuries were recorded during the quarter across 1.24 million cumulative man-hours.

VERIFIED PARAMETER REGISTRATION:

â€¢ **Quarterly Coal Extraction:**	1.48 MT	101.4% of quarterly plan
â€¢ **Quarterly Overburden Moved:**	4.22 M.Cu.m	Stripping ratio 2.85 m3/T
â€¢ **Lost Time Injury Frequency (LTIFR):**	0.00	Target < 0.20

SECTION 3.0 â€” PIT GEOMETRY & BENCH DESIGN PARAMETERS

Official Technical Record & Geological Data Logging

The opencast pit spans 2.4 km along strike and 1.1 km across dip. Current maximum pit depth is 142m from surface collar elevation (RL 298m).

Standard Pit Bench Design Dimensions

Bench Classification	Bench Height (m)	Bench Width (m)	Face Slope Angle	Safety Berm
Overburden Topsoil	6.0 m	18.0 m	45 deg Slope	3.0 m
Upper Sandstone Bench	12.0 m	28.0 m	70 deg Slope	5.0 m
Seam IV Coal Bench	8.5 m	25.0 m	75 deg Slope	4.0 m
Lower Sandstone Bench	14.0 m	32.0 m	70 deg Slope	6.0 m
Seam V Main Pit Floor	9.0 m	30.0 m	75 deg Slope	5.0 m

SECTION 4.0 â€” SEAM EXTRACTION SEQUENCE & MINING METHOD

Official Technical Record & Geological Data Logging

Extraction proceeds top-down utilizing a conventional shovel-dumper configuration combined with continuous haul road maintenance to maximize truck cycle speeds.

VERIFIED PARAMETER REGISTRATION:

â€¢ **Primary Extraction Seam:**	Seam IV (Thickness: 8.5m)	Grade G-4
â€¢ **Secondary Extraction Seam:**	Seam V (Thickness: 9.0m)	Grade G-5
â€¢ **Working Face Length:**	640 meters	Dual shovel advance

SECTION 5.0 â€” SLOPE STABILITY RADAR (SSR) MONITORING

Official Technical Record & Geological Data Logging

GroundProbe SSR-XT radar system operates 24/7 scanning the northern highwall face with sub-millimeter displacement detection sensitivity.

Highwall Slope Stability Velocity Thresholds

Alert Level	Velocity Threshold	Action Protocol	Current Reading
Green (Normal)	< 2.0 mm/day	Routine production logging	0.4 mm/day (Stable)
Yellow (Advisory)	2.0 - 5.0 mm/day	Inspect crest tension cracks	Inactive
Amber (Warning)	5.0 - 10.0 mm/day	Restrict personnel access	Inactive
Red (Critical)	> 10.0 mm/day	Immediate pit evacuation	Inactive

SECTION 6.0 â€” HAUL ROAD ENGINEERING & GRADIENT AUDITS

Official Technical Record & Geological Data Logging

The primary trunk haul road is 35m wide with a maximum ruling gradient of 1 in 16 (6.25%), surfaced with compacted crushed sandstone roadbase.

VERIFIED PARAMETER REGISTRATION:

â€¢ Trunk Road Width:	`35.0 meters`	3.5x widest truck width
â€¢ Ruling Gradient:	`1 in 16 (6.25%)`	DGMS Standard 1 in 16
â€¢ Safety Berm Height:	`2.1 meters`	Equal to dumper axle height

SECTION 7.0 â€” SHOVEL-DUMPER FLEET CYCLE TIME OPTIMIZATION

Official Technical Record & Geological Data Logging

Telemetric dispatch tracking registered an average round-trip cycle time of 16.4 minutes for the 100T dumper fleet over an average 2.8 km one-way haul distance.

Fleet Cycle Time Breakdown

Cycle Segment	Avg Duration (mins)	Benchmark (mins)	Efficiency %
Shovel Loading (4-pass)	2.8 mins	2.5 mins	89.2%
Loaded Haul to Dump	6.5 mins	6.2 mins	95.3%
Dumping & Positioning	1.4 mins	1.2 mins	85.7%
Empty Return Haul	4.8 mins	4.5 mins	93.7%
Queue & Spotting	0.9 mins	0.8 mins	88.8%

SECTION 8.0 â€” DRILLING & BLASTING OPTIMIZATION PARAMETERS

Official Technical Record & Geological Data Logging

Overburden blasting utilized 250mm diameter blast holes drilled with Atlas Copco DML rotary rigs on an 8m x 9m staggered pattern.

VERIFIED PARAMETER REGISTRATION:

â€¢ Specific Explosive Charge:	`0.68 kg / Cu.m`	Bulk emulsion matrix
â€¢ Average Powder Factor:	`1.47 Cu.m / kg`	Optimum fragmentation
â€¢ Mean Fragment Size (K50):	`0.28 meters`	Zero boulders > 1m

SECTION 9.0 â€” GROUND VIBRATION & AIR OVERPRESSURE MONITORING

Official Technical Record & Geological Data Logging

Peak Particle Velocity (PPV) was monitored at the boundary village (Kharwa) using Instantel Minimate Plus seismographs.

Seismograph Blast Vibration Log

Blast Event ID	Distance (m)	Dominant Freq (Hz)	Measured PPV (mm/s)	DGMS Limit
BLAST-CW-881	680 m	24.5 Hz	2.45 mm/s	10.0 mm/s
BLAST-CW-882	740 m	28.2 Hz	1.82 mm/s	10.0 mm/s
BLAST-CW-883	620 m	22.0 Hz	3.10 mm/s	10.0 mm/s
BLAST-CW-884	810 m	31.4 Hz	1.42 mm/s	10.0 mm/s

SECTION 10.0 â€” MINE DEWATERING & IN-PIT WATER MANAGEMENT

Official Technical Record & Geological Data Logging

Total sump capacity in the central dip floor stands at 450,000 m3 with a pumping battery capable of discharging 2,800 m3/hr to the surface settling ponds.

VERIFIED PARAMETER REGISTRATION:

â€¢ Main Sump Water Storage:	185,000 m3	41% capacity utilized
â€¢ High-Head Slurry Pumps:	4 x 750 kW Multi-Stage	Operational
â€¢ Sediment Settling Basin pH:	7.4	Neutral, TSS < 50 mg/L

SECTION 11.0 â€” COAL HANDLING PLANT (CHP) & SIZING OPERATIONS

Official Technical Record & Geological Data Logging

Raw coal from pit is tipped into 350-tonne receiving hoppers and processed through double-roll primary crushers to achieve uniform -100mm power station product.

CHP Screening and Crushing Throughput

CHP Stream	Rated Cap (TPH)	Actual Cap (TPH)	Availability %	Oversize %
Stream A (Primary)	1,500 TPH	1,420 TPH	92.4%	< 2.0%
Stream B (Secondary)	1,500 TPH	1,390 TPH	90.8%	< 2.0%
Rapid Loading Silo	3,000 TPH	2,850 TPH	96.2%	0.0%

SECTION 12.0 â€” IN-PIT CRUSHING & CONVEYING (IPCC) FEASIBILITY

Official Technical Record & Geological Data Logging

Engineering feasibility study evaluates replacing 18 dumpers with a semi-mobile in-pit crusher and overland belt conveyor system starting FY 2026.

VERIFIED PARAMETER REGISTRATION:

â€¢ Estimated Capex Investment:	**INR 145 Crores**	Payback 3.2 years
â€¢ Projected Opex Reduction:	**INR 42 / Tonne**	Diesel elimination
â€¢ Carbon Abatement:	**18,500 Tonnes CO2e/yr**	Electric conveyor

SECTION 13.0 â€” OCCUPATIONAL SAFETY, HEALTH & TRAINING KPIS

Official Technical Record & Geological Data Logging

Over 2,400 hours of simulator training for heavy vehicle operators and mandatory safety refresher courses were completed during the review period.

Quarterly Safety Audit Metrics

Safety Indicator	Achieved Score	Statutory Target	Verification
Fatal Accident Rate (FAR)	0.00	0.00	DGMS Certified
Near Miss Incidents Logged	42 Reported / Closed	100% Corrective Action	Internal Safety Cell
Dust Exposure Index	1.8 mg/m3	< 3.0 mg/m3	Occupational Hygiene Lab
Noise Level at Cabins	78 dB(A)	< 85 dB(A)	Acoustic Compliance

SECTION 14.0 â€” ENVIRONMENTAL DUST SUPPRESSION & TREE BELTS

Official Technical Record & Geological Data Logging

A network of 48 automated mist cannons and continuous water bowser spraying kept ambient PM10 concentrations below the 100 ug/m3 national standard.

VERIFIED PARAMETER REGISTRATION:

â€¢ **Water Bowsers Active:**	**8 x 28,000-liter Bowsers**	Continuous spraying
â€¢ **Fixed Mist Cannons:**	**16 Units at Haul Intersections**	Mist throw 60m
â€¢ **Green Barrier Tree Belt:**	**12,000 Acacia & Neem Trees**	Along pit boundary

SECTION 15.0 â€” MONTHLY PRODUCTION MILESTONES & COST BREAKDOWN

Official Technical Record & Geological Data Logging

Composite cost of production stood at INR 1,180 per tonne of coal, achieving high operational margins against market dispatch realization.

Cost of Coal Extraction Breakdown

Cost Component	Cost per Tonne (INR)	Percentage Share	YoY Variance
Heavy Machinery Fuel & Power	INR 342.00	29.0%	-4.2% (Fuel saving)
Explosives & Blasting	INR 112.50	9.5%	-1.8%
Workforce & Manpower	INR 415.00	35.2%	+3.1% (Annual DA)
Maintenance & Spares	INR 185.50	15.7%	-2.5%
Statutory Royalties & Cess	INR 125.00	10.6%	Fixed Statutory
TOTAL PRODUCTION COST	INR 1,180.00 / T	100.0%	-1.2% Overall

SECTION 16.0 â€” SHIFT-WISE DISPATCH LOG & SIDING LOADING EFFICIENCY

Official Technical Record & Geological Data Logging

Shift-wise analysis confirms highest loading rates during Night Shift (Shift C) due to cooler ambient conveyor operating temperatures.

Shift Loading Performance Average

Shift	Hours	Coal Dispatched (T)	Average Rake Load Time
Shift A (Morning)	06:00 - 14:00	5,420 Tonnes	2.9 Hours
Shift B (Evening)	14:00 - 22:00	5,180 Tonnes	3.1 Hours
Shift C (Night)	22:00 - 06:00	5,890 Tonnes	2.7 Hours

SECTION 17.0 â€” ACTION ITEMS & ENGINEERING DIRECTIVES

Official Technical Record & Geological Data Logging

1. Complete road surfacing of Secondary Haul Road #2 with compacted blast furnace slag by end of month.

2. Install telemetry fuel sensors on all contractor-operated auxiliary water tankers.

3. Initiate geotechnical core drilling on northern face to verify fault extension F-04.

SECTION 18.0 â€” MINE MANAGER OFFICIAL CERTIFICATION

Official Technical Record & Geological Data Logging

I hereby certify that the mining operations detailed in this report have been carried out strictly in compliance with the Mines Act 1952 and Coal Mines Regulations 2017.

CRITICAL TECHNICAL OBSERVATION & DIRECTIVE:

Digitally logged in Coal Intelligence System under ID: MIN-CWS-24.

VERIFIED PARAMETER REGISTRATION:

â€¢ **First Class Mine Manager:**	Er. Arunav Sen, B.Tech (Mining)	Cert. No. FCC-8812
â€¢ **Safety Officer Sign-off:**	K. S. Rathore, DGMS Cert.	Verified 19-Nov-2024

SECTION 19.0 â€” APPENDIX: MINE PIT SCHEMATICS & SURVEY PLAN

Official Technical Record & Geological Data Logging

Mine Plan Reference Drawing No. CW/MINE/PLAN/2024/Q4.

Surveyed and mapped by Certified Mine Surveyor using Trimble S7 Total Station.

Mine Survey Bench Marks

Bench Mark ID	Easting	Northing	Reduced Level (RL m)
BM-CW-01	384210.15	2648910.45	298.45 m MSL
BM-CW-02	385640.80	2648720.10	292.10 m MSL
BM-CW-03	384990.20	2647890.30	285.60 m MSL